

Function

Subject: C Language

Date: 11/01/2026

1. What is a Function?

⇒ A function is a named block of code that performs a specific task.

- It allows you to write a piece of logic once and reuse it wherever needed in the program.
- This helps keep your code clean, organized, easier to understand and manage.

Some Important Points :

- Function is a block of code, which has some name for identification.
- A C program can have any number of functions.

Ex:

function name → functionName()
Function body { //some code }

Function Definition

- Function names must be unique in a Program.
- Function definition vs Function Call

Function Definition:

```
fn()
{
    printf("Hello");
    printf("I am a function");
}
```

Function Call:

```
int main()
{
    fn();
}
```

Subject: _____

Date: ____/____/____

- Function is a way to implement modularization.
- Modularization is splitting up of a bigger task into several smaller sub-tasks to reduce the complexity of a problem.
- You can compile a C file without having main() function but cannot run.

————: Functions are of two types :————

1. Predefined Functions
2. User defined Functions

————: Important Points :————

1. You can define functions in any order.
2. Execution of program always begin from main() function.
3. Function executes only when it is called.
4. Operating System calls main().
5. Jab bhi koi function call hota hai, Sabse pehle woh RAM mein local hota hai.
6. Function Jab end hota hai, tab uski memory release hoti hai and Control wapis wahi jata hai, Jahan se call hua tha.

Benefits of Functions

1. Easy to Read
2. Reduce Complexity
3. Easy to modify
4. Easy to debug
5. Code reusability
6. Avoids rewriting
7. Better memory utilization.

Ways to define a function

1. Takes Nothing, Returns Nothing
2. Takes Something, Returns Nothing
3. Takes Nothing, Return Something
4. Takes Something, Return Something

1. Takes Nothing Return Nothing

```
int main()
```

```
{  
    add();
```

```
    return 0;
```

```
}
```

```
void add()
```

```
{  
    int a, b, c;
```

```
    printf("Enter two numbers: ");
```

Subject: _____

Date: ____/____/____

```
scanf("%d %d", &a, &b);  
c = a + b;  
printf("Sum is %d", c);  
}
```

2. ———: Take Something Return Nothing :————

```
#include <stdio.h>  
void add(int, int);
```

```
int main()  
{
```

```
    int x = 10, y = 20;  
    add(x, y);  
    return 0;  
}
```

```
void add(int a, int b)  
{
```

```
    int c;  
    c = a + b;  
    printf("Sum is %d", c);  
}
```


Subject: _____

8. —: Take Nothing . Return Something : —

```
#include <stdio.h>
int addl();
```

```
int main()
```

```
{
    int k;
```

```
    k = addl();
```

```
    printf("Sum is %d", k);
```

```
    return 0;
}
```

```
int addl()
```

```
{
    int a, b, c;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &a, &b);
```

```
    c = a + b;
```

```
    return c;
}
```

NOTE:- i) return Can only return Single value.
 ii) return chal gaya to Samyho function
 end ho gaya.

4. _____: Take Something Return Something :—

```
#include <stdio.h>
```

```
int add (int, int);
```

 → Function Declaration

```
int main()
```

* Actual Argument — add function ke andar jo x and y hai.

```
{  
    int x, x=10, y=20;
```

```
    x = add(x, y);
```

 → Function Call by value

```
    printf("Sum is %d", x);
```

```
    return 0;
```

```
}
```

 → Formal Argument

```
int add (int a, int b)
```

```
{
```

```
    int c;
```

```
    c = a + b;
```

```
    return c;
```

```
}
```

Function Definition

Some Important Points

1. Function ke andar jo variable banate hain use local variable bolte hain.
2. Mostly hum global variable ko avoid karte hain.
3. Compiler kisi bhi function ka matlab pahle se nahi janta hai, sirf keyword ka matlab janta hai. Har function Compiler ke liye variable ke jaisa hi naya sabad hai, to jaise variable ka declaration karte ho vaise hi function ka bhi declaration karna chahiye.
4. Variable ke Case me agar tum global variable banate ho to usko memory program ke pure life-time tak mili rehti hai, lekin function ko memory tabhi milti hai jab function call hoti hai. So aap function ko globally declaration kar sakte ho.
5. Function Prototype ko hi hum Function declaration bolte hain.
6. Call by value — Jab bhi koi function call karte ho aur call karte waqt usme koi argument pass karte ho to use Call by value bolte hain.
7. Header file me predefined function ka declaration hota hai, definition nahi.
8. Library file me Predefined function ka definition hota hai.